

LECTURE 3

Topics

Interest rates and inflation, Historical returns, Return statistics, Downside risk, Returns of long-term investments

- Recall from Lecture 1 that bank time deposits, such as checking or savings accounts, carry bank interest rates
- Each bank sets its own checking and savings interest rates to balance:
 - The bank's opportunities to generate revenues (e.g., extend loans to firms)
 - The bank's costs associated with compensating depositors for lending their capital to the bank
- However, bank rates are heavily influenced by the **Federal Fund Rate** (FFR), which is the rate at which banks lend to each other on the *interbank lending market*
- The Federal Reserve Banks (FRB) are the largest participant on the interbank lending market and they borrow/lend from banks to guide the FFR towards a *target* FFR set by FRB

- The FRB is the *Central Bank* of the US, which is basically the bank of the (commercial) banks
- Any central bank in the world has a *double mandate*:
 - Price stability
 - Full employment of the country's workforce
- FRB achieves this mandate by controlling the *supply of money* available to the public
 - When prices increase out of control, FRB reduces the supply of money (*contractionary* monetary policy)
 - When prices don't increase enough, FRB increases the supply of money (*expansionary* monetary policy)
- FRB controls the supply of money by targeting the FFR
- Higher target FFR $\Rightarrow$ banks want to lend to FRB $\Rightarrow$ banks offer higher savings rate to attract capital $\Rightarrow$ consumers spend less and save more $\Rightarrow$ money supply available to public goes down

- We refer to the FFR as *the interest rate* of the US economy, because it affects all bank interest rates
- It also affects the rates at which the US government, municipalities, and corporations borrow through loans and bonds
- Finally, it affects the rate of return that investors demand on all equity and debt securities
- While the interest rate is known now (because target FFR is known), it may change in the future
- If investors *expect* the interest rate to increase in the future, they expect a higher rate of return from *all* their investments in equity and debt securities
- Higher expected rates of return on securities increases the firms' cost of raising capital and thus their current stock prices fall

Effect of higher expected interest rates on current stock prices

Higher expected interest rates $\Rightarrow$ Higher expected rate of return on all investments in equity and debt securities $\Rightarrow$ Firms' cost of raising capital rises $\Rightarrow$ Current stock prices fall

Effect of lower expected interest rates on current stock prices

Lower expected interest rates $\Rightarrow$ Lower expected rate of return on all investments in equity and debt securities $\Rightarrow$ Firms' cost of raising capital falls $\Rightarrow$ Current stock prices rise

- The FFR interest rate that the FRB targets and the interest rates that banks set are sometimes referred to as *nominal* interest rate
- Nominal interest rates can be different from *real* interest rates:
 - Nominal interest rates are the growth rate of your money
 - Real interest rates are the growth rate of your purchasing power
- We can calculate the real interest rate, r_{Real} , from the nominal interest rate, r_{Nom} , using the Fischer equation:

$$1 + r_{Real} = \frac{1 + r_{Nom}}{1 + i}$$

or simply

$$r_{Real} \approx r_{Nom} - i$$

where i is the *inflation rate* that we discuss next

- The **inflation rate**, i , is calculated as the rate of change in the *Consumer Price Index* (CPI):

$$i = \frac{\text{Current CPI} - \text{Previous CPI}}{\text{Previous CPI}}$$

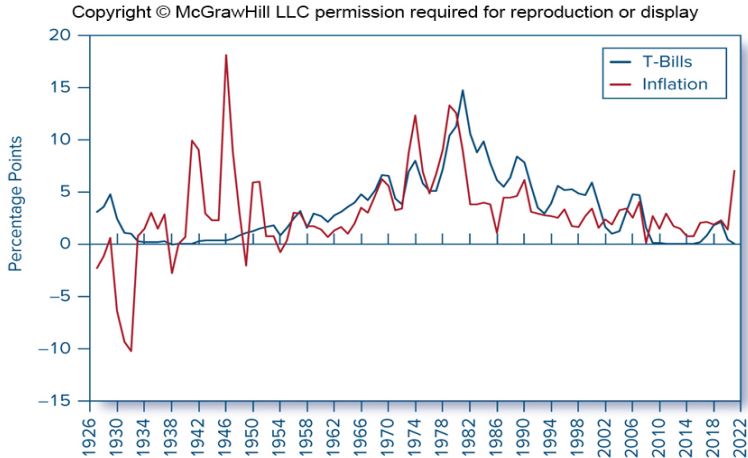
- CPI is basically the price (in relative terms) of a specific *basket* of consumer goods and services
- CPI tracks the *cost of living* for the US consumer
(<https://www.bls.gov/cpi/overview.htm>)
- For example, if inflation rate was 5% last year, then what you could buy with \$100 one year ago would require \$105 now

- The inflation rate calculated above is in the past, because it is based on historical CPI levels
- When the FRB decides on whether to maintain the existing target FFR or change it, they look at factors that affect current and future price stability and employment
- In particular, they look at future price stability, i.e., future changes in CPI
- While FRB cannot predict futures changes in CPI, they can forecast *expected* changes in CPI, i.e., *expected inflation rate*
- Similarly, the FRB forecasts the real interest rate (i.e., real FFR)
- Then the FRB should set the nominal target interest rate according to Fisher's equation:

$$r_{Nom} = r_{Real} + E(i)$$

where $E(i)$ is expected inflation

- Irvine Fisher hypothesized that **real interest rates** are completely unrelated to **nominal interest rates**
- Using the Fisher equation above, the hypothesis states that the FRB sets the nominal interests rates solely to offset the potential changes in expected inflation rate
- Whether this hypothesis is supported by the data or not is difficult to test because of the need to forecast expected inflation and real interest rates
- However, we can visually inspect whether the nominal interest rates and past inflation rates track each other over time



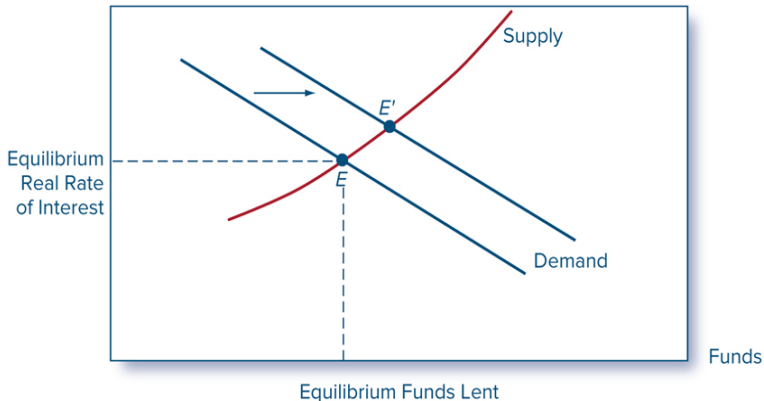
- In the plot above, T-Bills yields capture nominal interest rates
- The plot suggests that T-Bills rates and inflation rates seem to track each other relatively well post 1950's
- In the aftermath of World War II, the FRB set the FFR rates to a low target to keep the cost of rebuilding low
- In 1951, a US Treasury - FRB Accord effectively ended this policy and the FRB was given full control over the monetary policy and the FFR
- Thus, the plot captures well this change in monetary policy
- While we discussed how the FRB controls the nominal interest rate (FFR), what determines the real interest rate of the economy?

- A 2-year amortizing loan has a face value of \$1000 and a nominal interest rate of 5% per year
- Suppose that the annual expected inflation rate is 3%
- At the end of the first year the loan payment in nominal terms (i.e., in money) is \$537.80 ($=\text{PMT}(5\%,2,1000,0)$)
- However, the loan payment in real terms (i.e., in goods and services) is $\frac{\$537.80}{(1+3\%)} = 522.14$
- Similarly, the loan payment in real terms in the second year is $\frac{\$537.80}{(1+3\%)^2} = 506.93$
- The lender's purchasing power from the loan payment diminishes over time because of the inflation
- In contrast, the borrower's purchasing power increases because its loan payments are worth less in real terms
- The borrower's effective borrowing rate is the real rate of $2\% = 5\% - 3\%$

- We can employ a simple demand-supply framework for capital to understand real interest rates
- According to the above, lower real interest rates lead to higher demand for capital
- At the same time, lower real interest rates lead to less capital being provided by the lenders
- The supply and demand curves cross at a point known as *the equilibrium*
- Real interest rates are ultimately determined at such equilibrium points

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Real Interest Rate



- Demand/supply curves shift as the economy changes

- When investing in a financial asset over a certain time period ("holding period"), investors focus on the *rate of return* of that investment over the holding period
- The rate of return for an asset over a holding period of time T can be calculated with the formula:

$$r_T = \frac{P_T + d_T - P_0}{P_0}$$

- where:
 - P_0 = purchase price at the beginning of the holding period
 - P_T = selling price at the end of the holding period
 - d_T = income generated by the asset during holding period
- The difference $P_T - P_0$ is the investment's *capital gain*
- d_T are usually stock dividends or bond coupons

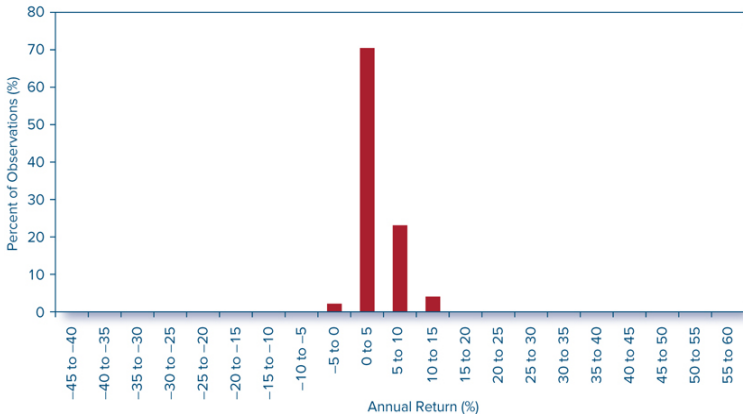
- IBM common stock pays regular quarterly dividends that are announced a few weeks before the scheduled payments
- Suppose you bought 100 IBM shares on February 1, 2024 for \$186.2 per share and sold them on Aug 20, 2024 for \$198.3 per share
- IBM paid a quarterly dividend of \$1.66 per share on February 9, \$1.67 per share on May 9, and \$1.67 per share on August 9
- What should be the return on your IBM investment over the holding period Feb 1 - Aug 20?
- What should be the annualized return of your investment?

- In finance we have plenty of (historical) data on all financial assets that are traded publically
- An important component of any investment analysis is to calculate the *historical* returns for each financial asset that you are considering and get a sense of:
 - Range of values (i.e., min and max)
 - Average value
 - Most likely value
 - Variability or Risk associated with these returns (i.e., how spread out are the historical returns)
- One convenient way to summarize all these characteristics is through a histogram or a historical frequency *distribution*
- We illustrate the historical distribution of several asset classes

● T-Bills

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Panel A: Treasury Bills

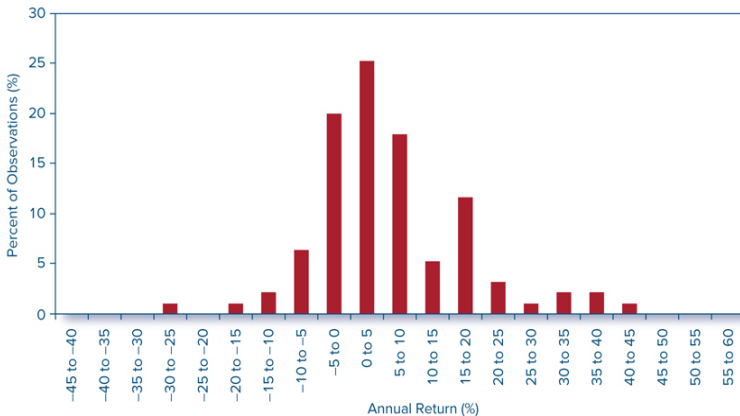


- Note the range is between -5% and 15% annualized returns, and the most likely return value is between 0% and 5%

- 30-year Treasury bonds

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Panel B: 30-Year Treasury Bonds

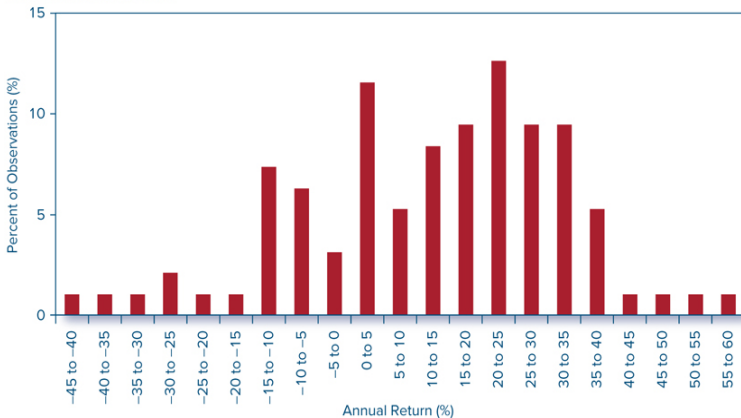


- Note the range is between -30% and 45% annualized returns, and the most likely return value is between 0% and 5%

- Common stocks

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Panel C: Common Stocks



- Note the range is between -45% and 60% annualized returns, and the most likely return value is between 20% and 25%

- While we were able to spot the range and most likely value directly from the historical distribution, calculating the average value and the variability require that we go back to the time-series of historical data
- To calculate average or *mean* historical return, $\bar{r}$, from a sample of length T , we average across all the observations in the sample:

$$\bar{r} = \frac{r_1 + r_2 + \dots + r_T}{T}$$

- To calculate the return variability or risk we use the standard deviation of returns (or "return volatility") that is typically denoted with σ

$$\sigma = \sqrt{\frac{(r_1 - \bar{r})^2 + (r_2 - \bar{r})^2 + \dots + (r_T - \bar{r})^2}{T^*}}$$

- $T^* = T$ usually, but when statistical rigor is required, $T^* = T - 1$
- For the most part we will calculate these statistics using Excel

- Statistics for the distributions of T-Bills, T-Bonds and Stocks that we analyzed above:

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Table 5.4

Risk and return of investments in major asset classes; estimates from annual data, 1927–2021

	T-Bills	T-Bonds	Stocks
Average	3.30%	5.96%	12.17%
Risk premium	N/A	2.66	8.87
Standard deviation	3.10	11.58	19.89
Max	14.71	41.6803	57.35
Min	-0.02	-25.96	-44.04

- The risk premium of an asset is the difference between the mean return of the asset, $\bar{r}$, and the current T-Bills yield, $r_{\text{T-Bill}}$:

$$\text{Risk premium} = \bar{r} - r_{\text{T-Bill}}$$

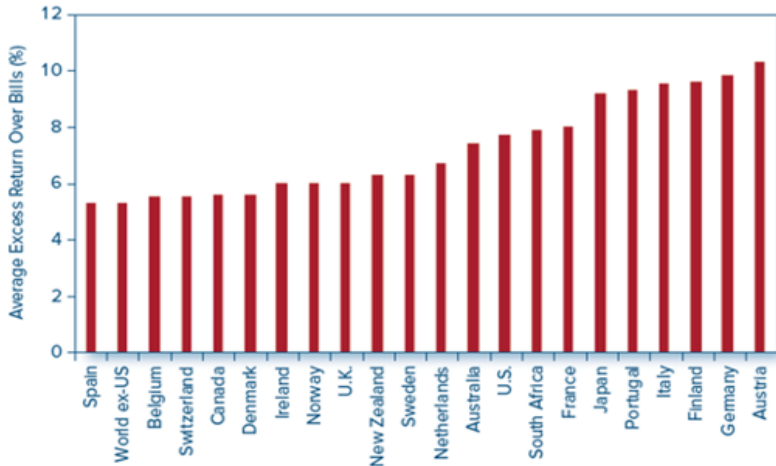
- Return volatility measures how much variability there was in returns, historically
- This variability is an indication of the unpredictable swings in future returns, or the *total risk* of the returns
- Thus, return volatility measures total risk in returns
- The risk premium of an asset capture how much compensation an investor receives, historically, from bearing the risk associated with that asset relative to the T-Bills
- From the table above, we notice an intuitive pattern: riskier asset classes tend to compensate investors more
- Whether this pattern generalizes to all asset classes is something we will investigate later in the class

- In the pattern above we noticed that stocks provide more compensation relative to T-Bills or bonds but have much higher levels of risk
- William Sharpe argues that if we normalize the compensation by dividing with the return volatility we obtain a measure of compensation per unit of risk
- Specifically, he proposed the so-called Sharpe's Ratio:

$$SR = \frac{\bar{r} - r_{T-Bill}}{\sigma}$$

- In certain situations, the Sharpe's ratio can be used to compare the performance of different well-diversified portfolios
- We can also use it to get a sense of how big the return compensation per unit of risk is across different markets

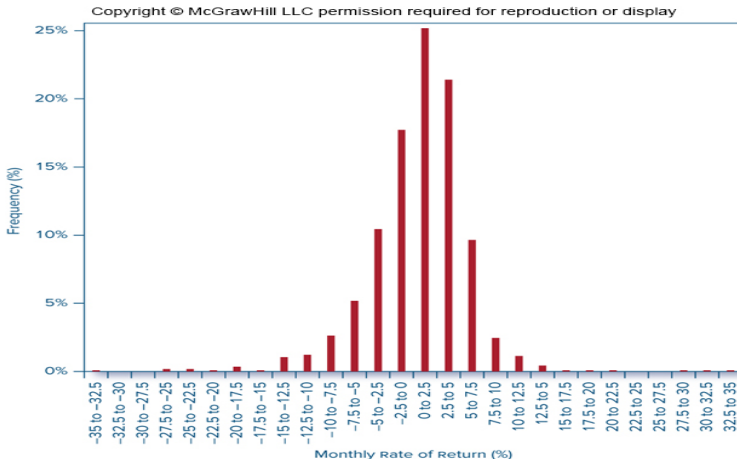
Panel A: Risk Premia



Panel B: Sharpe Ratios



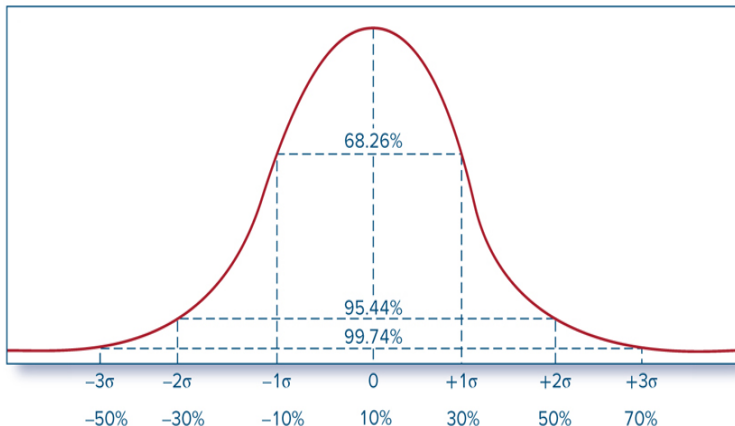
- If we change from annual returns to monthly returns, the return distribution changes as well, and sometimes quite significantly
- For example, here is the distribution of monthly stock returns



- We note that the distribution of monthly stock returns has bell-like shape
- We associate this shape with one of the most common type of distributions in statistics, namely the *normal distribution*
- The normal distribution is central to the theory and practice of investment analysis
- Some of the features that makes it easy to work with:
 - It is characterized by two parameters only, the mean and the standard deviation
 - It is symmetric, and the mean and the most-likely value are the same
 - 99.74% of the values are within 3 units of standard deviation from the mean

- For example, here is the normal distribution with mean 10% and standard deviation 20%

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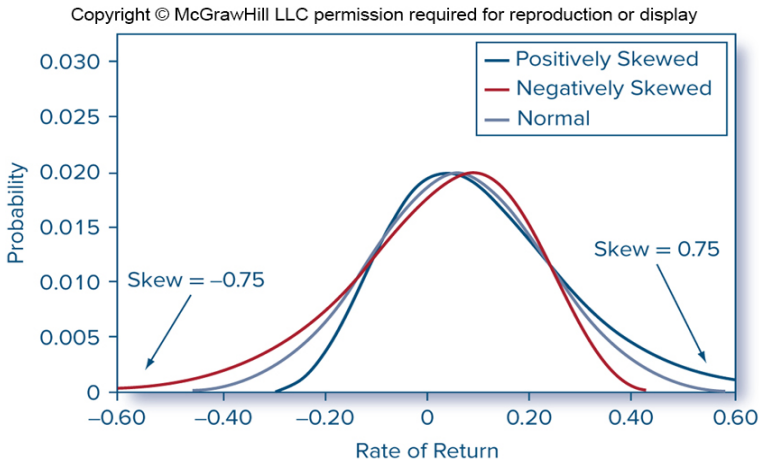
- An outcome is considered *extreme* if it is more than $3 \times$ units of standard deviation from the mean
- Extreme events are very rare in a normal distribution (99.74% of the range of values are within 3 units of standard deviation)
- Of course, historical distributions deviate in various ways from a normal distribution
- To measure the extent of the deviation, we look at:
 - Skewness = deviations from symmetry

$$\text{Skewness} = \frac{(r_1 - \bar{r})^3 + (r_2 - \bar{r})^3 + \dots + (r_T - \bar{r})^3}{T^* \times \sigma^3}$$

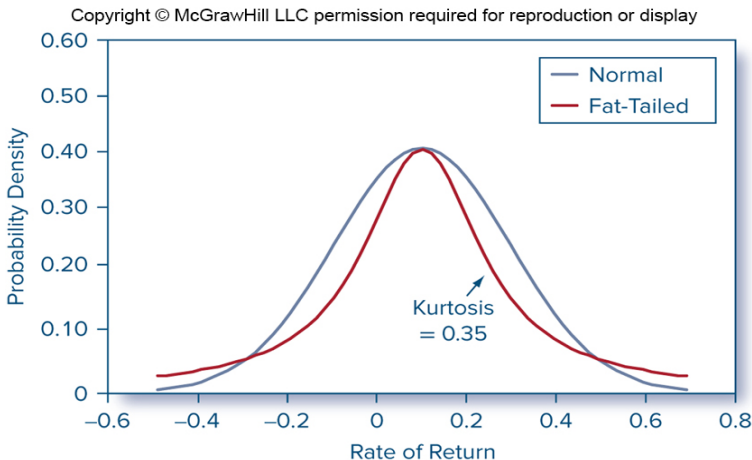
- Kurtosis = the frequency of occurrence of extreme events

$$\text{Kurtosis} = \frac{(r_1 - \bar{r})^4 + (r_2 - \bar{r})^4 + \dots + (r_T - \bar{r})^4}{T^* \times \sigma^4} - 3$$

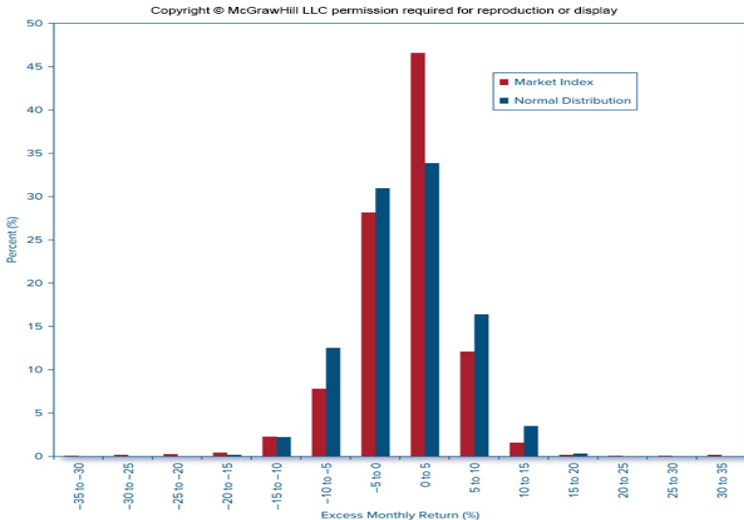
- The plot below illustrates positive versus negative skewness relative to a normal distribution (which has skewness = 0 because of symmetry)



- The plot below illustrates non-zero kurtosis relative to a normal distribution (which has kurtosis = 0 because it has almost no extreme events)



- Another way to analyze deviations from normal distribution is to "fit" a normal distribution curve over the original distribution



- One type of deviation from the normal distribution that is very useful is the frequency of extreme events (3-sigma events)
- The higher this frequency is, the "fatter" the tails of the distribution
- *Downside risk* is a term used to capture the frequency of extreme events on the negative side (3-sigma events to the left of the mean)
- Downside risk can be measured with:
 - Value-at-risk ("VaR")
 - Expected shortfall
 - Lower partial standard deviation
 - Actual frequency of 3-sigma events
- All these measures are relatively technical, so we focus on the most common of them, namely the VaR

- You invested in a portfolio of assets and you are worried about the size of possible future losses
- You are aware of the historical return distribution for your investment, and remember that the extreme negative returns are to the left of the mean return
- Value-at-Risk is the negative return value in the distribution that indicates where the area of most negative extreme returns starts
- This area is usually set to the 5% most extreme events
- The potential loss is then calculated with the formula:

$$\text{VaR Loss} = \text{VaR Value} \times \text{Current net worth of your investment}$$

- To calculate the VaR value, we simulate returns from the original historical distribution, and then we check which of the simulated returns fall in the area of extreme returns

- The historical return distribution of an index that tracks the stock market reveals valuable information about:
 - Possible scenarios for the return of the market in the future
 - The likelihood or probability associated with each scenario
- For example, the plot on the distribution of monthly stock returns that we discussed above reveals:
 - Scenario 1: Less than -10% [Probability = 3%]
 - Scenario 2: Between -10% and -5% [Probability = 7.5%]
 - Scenario 3: Between -5% and 0% [Probability = 27%]
 - Scenario 4: Between 0% and 5% [Probability = 49.5%]
 - Scenario 5: Between 5% and 10% [Probability = 12%]
 - Scenario 6: More than 10% [Probability = 2%]
- The collection of return outcomes for each scenario and the associated probability is called a *probability distribution*

- Probability distributions attempt to provide an accurate characterization of an asset's possible future return outcomes (the scenarios) and their corresponding probabilities
- It usually involves a lot more analysis than our simple mapping from historical distribution above
- Investor use probability distributions to get a more accurate read on the *expected* rate of return of an asset as well as the return volatility of the asset
- We illustrate below how these statistics can be calculated from a probability distribution

- Suppose that it is known that the probability distribution for the monthly market return is as follows:
 - Scenario 1: $r_1 = -10\%$ with probability $p_1 = 15\%$
 - Scenario 2: $r_2 = -5\%$ with probability $p_2 = 20\%$
 - Scenario 3: $r_3 = 0\%$ with probability $p_3 = 17\%$
 - Scenario 4: $r_4 = 5\%$ with probability $p_4 = 30\%$
 - Scenario 5: $r_5 = 10\%$ with probability $p_5 = 18\%$
- Then the expected market return for the next month, $E(r)$, can be estimated to be

$$E(r) = r_1 \times p_1 + r_2 \times p_2 + \dots + r_N \times p_N$$

$$E(r) = -10\% \times 15\% + -5\% \times 20\% + 0\% \times 17\% + 5\% \times 30\% + 10\% \times 18\%$$

$$E(r) = 0.8\%$$

- The market return volatility for the next month, σ , can be estimated to be

$$\sigma = \sqrt{(r_1 - E(r))^2 \times p_1 + (r_2 - E(r))^2 \times p_2 + \dots + (r_N - E(r))^2 \times p_N}$$

$$\sigma = \sqrt{(-10\% - 0.8\%)^2 \times 15\% + \dots + (10\% - 0.8\%)^2 \times 18\%}$$

$$\sigma = 6.7\%$$

- An important result in econometrics is that, with a long enough time-series, the expected return and the mean historical return (calculated earlier) are almost indistinguishable
- Return volatility can also be estimated more and more precise from historical data when the frequency of the data is higher
- For the rest of the class, we will rely on these results to estimate future expected return and return volatility from historical data

- Recall that the historical distribution of monthly stock returns resembles closely a normal distribution
- We say that monthly returns are close to being normally distributed
- If we were to invest for T months and earn returns of r_1 for month 1, r_2 for month 2, and so on, then the holding return for the investment, r_* , can be calculated:

$$1 + r_* = (1 + r_1) \times (1 + r_2) \times \dots \times (1 + r_T)$$

- Effectively, the return earned in one month is compounded for the rest of the investment period
- The compounding makes r_* has a distribution that is very different from the normal distribution

- We illustrate with an example
- Suppose you invest initial wealth of $w_0 = \$100$ in an asset that has an expected monthly return of 1%
- The actual return of the investment is risky and it could be either 3% or -1% with equal probability (i.e., 50%)
- In particular, we can confirm that the expected monthly return of the asset is 1%:

$$E(r) = 3\% \times 50\% + -1\% \times 50\% = 1\%$$

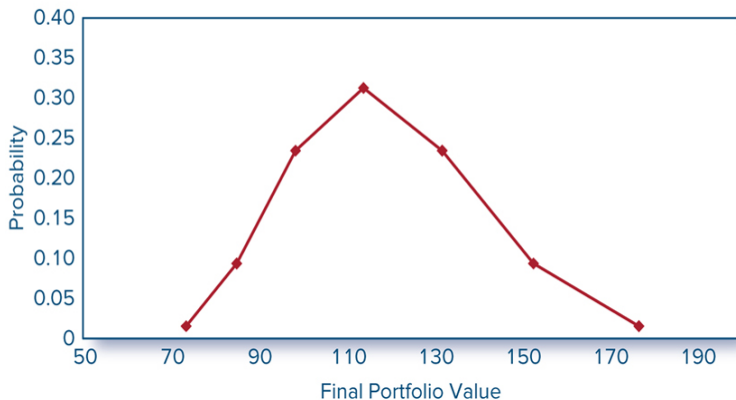
- We can plot the *terminal* wealth from your investment, w_T , after $T = 6, 20$, and 40 months, respectively
- We calculate the terminal wealth as follows:

$$w_T = w_0 \times (1 + r_*)$$

- The distribution of the terminal wealth after $T = 6$ months

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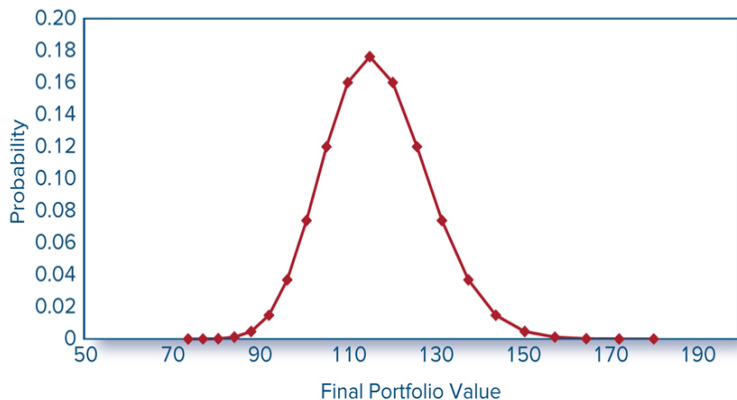
Panel A: 6 months



- The distribution of the terminal wealth after $T = 20$ months

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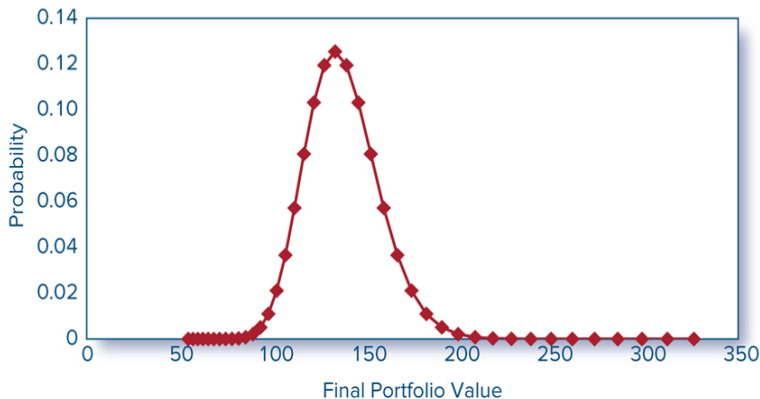
Panel B: 20 months



- The distribution of the terminal wealth after $T = 40$ months

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Panel C: 40 months



- Note that as the investment horizon increases from $T = 6$ to 40 months, the distribution of the terminal wealth is bell-shaped like the normal distribution, but the right tail is much longer than the left
- The longer right tail is the result of compounding which has the effect of significantly increasing the frequency of extreme positive returns
- The asymmetry in the right and left tails means that the distribution is no longer normal
- Instead, this type of distribution is closer to something called *lognormal* distribution
- If monthly returns, r , have normal distribution, then the exponential e^r will have a lognormal distribution

- Another way to see why the terminal wealth has a lognormal distribution relies on the following approximation for small values of r :

$$1 + r \approx e^r$$

- Then, we can rewrite the holding return above as follows:

$$e^{r_*} = e^{r_1} \times e^{r_2} \times \dots e^{r_T}$$

$$e^{r_*} = e^{r_1 + r_2 + \dots + r_T}$$

- Because normal distribution is additive, $r_1 + r_2 + \dots + r_T$ has a normal distribution
- As a result, the terminal wealth will have a lognormal distribution because

$$W_T = W_0 \times e^{r_1 + r_2 + \dots + r_T}$$